1. Model Architectures

Four distinct neural architectures were implemented to classify news articles as True or Fake

- **RNN & LSTM**: Sequential baselines using a 32-dimensional embedding and 64 hidden units.
- **Custom Transformer**: Manually implemented architecture featuring token embeddings, sine/cosine positional encodings, and a TransformerEncoder with mean pooling.
- **Fine-Tuned BERT**: Pretrained ber-base-uncased model adapted for sequence classification.

2. Results

```
=== RNN Results ===
Accuracy: 59.51%
Precision: 0.61
Recall: 0.64
F1 Score: 0.63
Mean Inf Speed: 0.000100s/sample
Total Parameters: 70,273

=== LSTM Results ===
Accuracy: 61.47%
Precision: 0.70
Recall: 0.47
F1 Score: 0.56
Mean Inf Speed: 0.000117s/sample
Total Parameters: 89,089

=== Custom Transformer Results ===
Accuracy: 98.26%
Precision: 1.00
Recall: 0.97
F1 Score: 0.98
Mean Inf Speed: 0.000862s/sample
Total Parameters: 690,241
```

```
=== Fine-Tuned BERT Results ===
Accuracy: 99.82%
Precision: 1.00
Recall: 1.00
F1 Score: 1.00
Mean Inf Speed: 0.048630s/sample
Total Parameters: 109,483,009

--- Study 1: Positional Encoding Impact ---

=== Transformer (With PE) Results ===
Accuracy: 98.22%
Precision: 1.00
Recall: 0.97
F1 Score: 0.98
Mean Inf Speed: 0.000849s/sample
Total Parameters: 690,241

=== Transformer (No PE) Results ===
Accuracy: 98.17%
Precision: 1.00
Recall: 0.97
F1 Score: 0.98
Mean Inf Speed: 0.000840s/sample
Total Parameters: 690,241

--- Study 2: Attention Heads (2 vs 8) ---

=== Transformer (2 Heads) Results ===
Accuracy: 97.77%
Precision: 0.99
Recall: 0.97
F1 Score: 0.98
Mean Inf Speed: 0.000586s/sample
Total Parameters: 690,241

=== Transformer (8 Heads) Results ===
Accuracy: 98.26%
Precision: 1.00
Recall: 0.97
F1 Score: 0.98
Mean Inf Speed: 0.000730s/sample
Total Parameters: 690,241
```

Model	Accuracy	Precision	Recall
RNN	59.51%	0.61	0.64
LSTM	61.47%	0.70	0.47
Custom Transformer	98.26%	1.00	0.97
Fine-Tunes BERT	99.82%	1.00	1.00

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3. Discussion

The Fine-Tuned BERT model achieved the highest performance. THis is because of the massive parameter count and extensive pretraining, which provided deeper understanding of linguistic context far beyond what smaller models can learn from the task-specific data alone.

Custom Transformer is significantly faster at inference, BERT's performance makes it better choice where accuracy is essential.

4. Design Studies

Study 1: Positional Encoding Impact

- **With PE**: 98.22% Accuracy
- **No PE:** 98.17%
- **Observation:** In the specific dataset, the impact of PE was marginal. Thus suggests of high-signal keywords that attention mechanism is able to identify regardless of their exact position in the sequence

Study 2: Attention Heads

- **2 Heads:** 97.77% Accuracy
- **8 Heads:** 98.26% Accuracy
- **Observation:** Increasing the number of attention heads improved performance, as it allows the model to simultaneously attend to different representation subspaces.